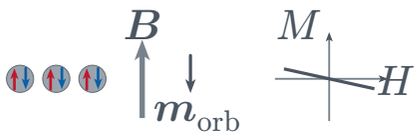
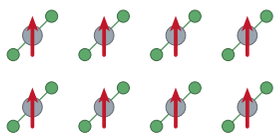
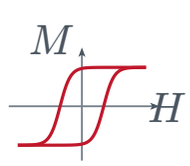
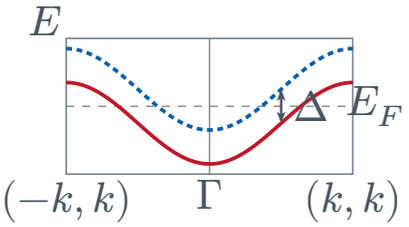
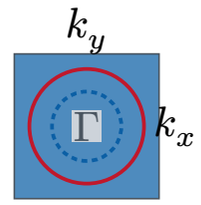
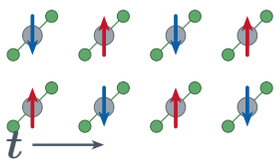
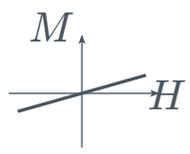
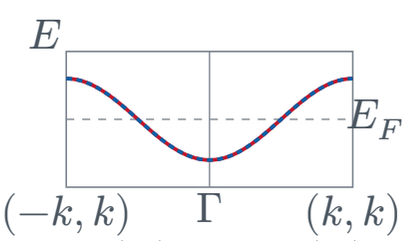
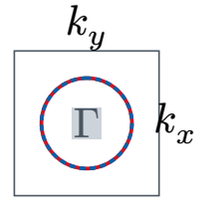
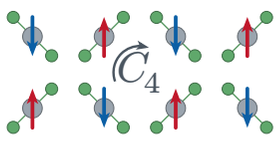
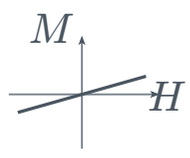
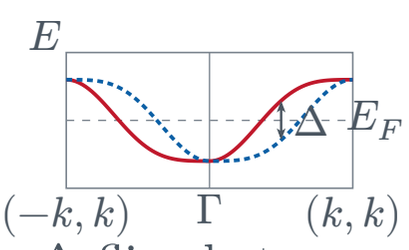
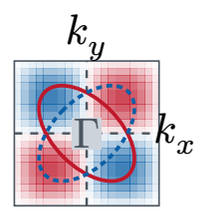
all three rows are collinear; $M(H)$ singles out the ferromagnet, and only the spin splitting $\Delta(\boldsymbol{k}) = E_\uparrow(\boldsymbol{k}) - E_\downarrow(\boldsymbol{k})$ separates the other two

	real-space structure	**response $M(H)$**	**spin-resolved bands**	**spin splitting $\Delta(\boldsymbol{k})$**
Diamagnetism	 closed shells, $\boldsymbol{m}_{\text{orb}}$ opposes $\boldsymbol{B}$ $\chi \sim -10^{-5} < 0$			
Ferromagnetism	one sublattice, every moment parallel	 identical ligand axes	 hysteretic, $M_r \neq 0$	 rigid s-wave split, same sign at every $\boldsymbol{k}$	 concentric contours, majority larger
Paramagnetism	 disordered moments, $\langle \boldsymbol{m} \rangle = 0$ $\chi = C/T \sim 10^{-3} > 0$			
Antiferromagnetism	sublattices related by a translation $\boldsymbol{t}$, or by inversion	 identical ligand axes	 linear, no remanence	 $E_\uparrow(\boldsymbol{k}) = E_\downarrow(\boldsymbol{k})$ everywhere	 one doubly degenerate contour
Ferrimagnetism	 antiparallel but $\lvert\boldsymbol{m}_A\rvert \neq \lvert\boldsymbol{m}_B\rvert$, so $\lvert\boldsymbol{M}_A + \boldsymbol{M}_B\rvert \neq 0$			
Altermagnetism	sublattices related only by a C_4 rotation	 ligand axes rotated 90°	 identical to the antiferromagnet	 Δ flips between the C_4-related diagonals	 $\Delta \propto \sin k_x \sin k_y$, nodal on axes

$E_\uparrow$ $E_\downarrow$ curves drawn on top of each other are spin degenerate $\Delta > 0$ $\Delta < 0$

non-relativistic limit throughout; $M(H)$ axes not to scale across panels; d-wave altermagnet shown, g- and i-wave also exist